Neural Dynamics of Narrative Structure-Building in Visual Stories

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Abstract

While models of discourse comprehension describe the process of structure building during mental model construction, neurophysiological explorations of this process are limited. Here, we use time-frequency analysis of EEG data to explore the spectral power dynamics associated with the narrative comprehension of comics. Using an existing dataset wherein 22 participants viewed sets of six sequential comic panels, we performed spectral decomposition from theta to gamma bands over the full extent of narrative processing (10+ seconds). Power incrementally decreased in both alpha (8 -12 Hz) and low beta (12.5-20 Hz) frequency bands as narratives unfolded. These results are contextualized in the literature, where some suggest that alpha and low beta frequency bands act as mechanisms of suppression and enhancement to modulate attention. Study findings are consistent with changes in alpha and low beta power reflecting domain-general processes of narrative structure building during discourse comprehension.

Keywords: narratives; structure building; time-frequency; EEG; alpha; beta

1. Introduction

How does the brain comprehend coherent sequences of meaningful information? Despite the ubiquity of talk and text in our daily experience with stories, story comprehension has been posited to use similar cognitive processes regardless of whether the story is told through words or pictures. In the work that follows, we use time-frequency analysis (TFA) of electroencephalography (EEG) data to explore narrative structure building in the context of sequential images that convey coherent stories – or, more informally, comics. While theories of narrative comprehension converge on some core cognitive elements, there remains a gap in our understanding of the neural mechanisms underlying those processes.

We first define several key terms to clarify our theoretical stance and establish a common interpretive framework. Importantly, we conceptualize *language* as not being restricted to written, spoken, or signed modalities but also encompassing systematically patterned meaningful information, organized by complex combinatorial systems, in other modalities (Cohn & Schilperoord, 2022). Just as words or text are the representational components of spoken or written language, visual elements within complex sequential images can comprise a "visual language" for communicating ideas or stories. Experimental work has shown that similar processes are employed during symbolic information processing no matter whether segmented units are verbal, like words, or visual, as are pictures (e.g., Cohn, 2020a).

We define discourse as the temporally organized use of language (whether verbal or visual) to express coherent ideas that span multiple propositions. This definition of discourse aligns with neurocognitive theories that highlight the relative complexity inherent in larger units of connected communication. We specifically focus on narratives, which belong to a particular genre of discourse-level communication. Although narrative theory has a long history with many different definitions (e.g., Bruner, 1991; Bruner, 1997; Richardson, 2000; Walsh, 2007), we define *narrative* or *story* according to commonalities among these approaches: a narrative consists of one or more actors with goals and motivations (Bruner, 1997) and a temporal sequence of causally connected events (Bruner, 1991; Richardson, 2000).

In the current study we use sequential images to explore discourse and narrative processing in the graphic modality; however, the cognitive mechanisms we describe are theorized to be similar across modalities. Indeed, models of narrative comprehension have often incorporated visual stories to argue that higher level cognitive processes are not specialized for any particular stimulus modality (Gernsbacher, 1990; Mandler & Johnson, 1977; Rumelhart, 1975; van Dijk & Kintsch, 1983). This project ultimately integrates research across disciplines to support a robust and generalizable account of the neurological correlates of narrative comprehension, as framed by the Structure Building model.

1.1. Models of Narrative Comprehension

Morton Gernsbacher's *Structure Building Framework* (1997) outlines three component processes for creating a coherent mental representation of any narrative: building a structure, integrating details into that structure, and shifting structures when confronted with incongruent information. The initial structure building step is *laying a foundation*: establishing contextual elements such as who the characters are, what they are doing, and their physical environment.

The framework also specifies ongoing evaluation of representational coherence while

constructing and maintaining a mental model (Gernsbacher, 1997). Once narrative themes or relationships have been established, the integration of new information depends on cognitive *suppression* and *enhancement* based on its narrative relevance. When semantic propositions are congruent with the narrative contour, their connections to the discourse structure should be enhanced, thus *mapping* them onto an existing structure. Mapping improves cognitive efficiency and strengthens the integrity of emerging narrative structures. In contrast, incoming information is less coherent with the current narrative structure at topical, temporal, or spatial boundaries; this means comprehenders must *shift* and initiate a new narrative substructure (Gernsbacher, 1990).

The temporal unfolding of discourse is not merely sequential, but a feature inherent to the emergent creation of meaning. The incremental and hierarchical nature of story organization yields a mental model with a cohesive structure that cannot be ascribed to the additive effects of constituent details alone. Often described as a *situation model* (van Dijk & Kintsch, 1983), this connectivity matrix includes granular information, semantic propositions, global inferences, and their interconnections which have been reinforced, suppressed or pruned through integration into the discourse context (Kintsch, 1988).

Theoretical models of narrative comprehension must account for cognitive processes and representations of information at a variety of levels. *Microstructural* discourse components are the most elementary units of meaning, and can be understood at a relatively local level (van Dijk, 1980). Structures at the word, phrase, and morphosyntactic levels are considered microstructural units of verbal communication (Coelho et al., 2006; van Dijk, 1980) while visual languages might include elements representing characters or key items in the environment, such as a bed or telephone (Cohn, 2013). Discourse *macrostructure* pertains to semantic elements and relationships that span microstructural boundaries, linking information in a coherent way that develops a thematic thread across narrative time (van Dijk, 1980).

While the Structure Building Framework has particularly influenced subsequent research on *text* comprehension, Gernsbacher stresses the parsimony of a domain-general model of cognitive comprehension. Indeed, even the original experiments grounding this framework were based on graphic narrative stimuli (Gernsbacher, 1985), resulting in a broad theory meant to apply to human cognition across sensory domains and functional contexts. Recent theories also stress the view that narrative comprehension employs domain-general processes beginning with the apprehension of information microstructure up to the level of macrostructural integration (Cohn, 2020b; Cohn & Schilperoord, 2022a; Loschky et al., 2020).

The *Scene Perception and Event Comprehension Theory (SPECT)* model (Loschky et al., 2020) describes the interplay between perceptual processes and event model construction during *visual* narrative processing. The SPECT model specifies front-end processes that extract information from the visual scene during single eye fixations, and back-end processes, based in executive control and memory, that integrate this perceptual information to create and maintain event models. The *Parallel Interfacing Narrative-Semantics Model* (Cohn, 2020b) also models key aspects of sequential image processing while explicitly specifying domain general cognitive mechanisms. Like the Structure Building Framework, these models of visual narrative comprehension emphasize bidirectional influences on detail appraisal as the situation model is continually updated.

1.2 Cognitive Processes Involved in Structure Building

The process of narrative structure building is thought to draw on a variety of cognitive control mechanisms. The perceptual integration of information at different levels of granularity is simultaneously modulated by two complementary facets of attentional control (Saghravanian, 2025). *Bottom-up* cognitive processes, as we define them here, refer to temporally, spatially, or hierarchically local influence (Rauss & Pourtois, 2013; Saghravanian, 2025). During narrative comprehension, bottom-up processes combine adjacent bits of microstructural information into more complex semantic structures. For example, hearing the introduction to a story might entail the assembly of each word in the sentence "A bear lived in the forest" together to comprise an idea (proposition) about a furry animal in a wooded area. Appraising a visual scene with straight lines similar to stream banks, triangular shapes that evoke conifers, and a furry looking four-legged animal might also cohere into a propositional forest.

In contrast, *top-down* processes influence the interpretation of information based on more global, longer-range, or "higher order" elements such as multi-modal models, predictions, or past experience. These executive functions are comprised of three main constructs: inhibition, updating (or working memory), and shifting (Karr et al., 2018; Niendam et al., 2012), which map fairly well onto the main components of Gernsbacher's structure building. More complex models of executive function posit distinctions between composite and sub-constructs (Friedman & Miyake, 2017) that can usefully describe attentional control at a task-specific level. For instance, to compare each new semantic element to the narrative structure, comprehenders must use both relational (Rubin & Cohen, 2017) and working memory (Lê et al., 2023; Lê et al., 2012) to hold and manipulate multiple substructures in mind simultaneously. Selective attention is required to maintain focus on discourse relevant information (Jacoby & Fedorenko, 2020; Lesenfants & Francart, 2020), and conflict or surprisal monitoring is needed to identify propositions destined to become new sub-structures (Federmeier et al., 2020; Rommers et al., 2017).

In the context of perceptual integration during comprehension, bidirectional exchange of information across levels of discourse structure helps us build, refine, and adjust mental models based on both local and global relationships (Federmeier et al., 2020; Kuperberg & and Jaeger, 2016). Expectations that bears do tend to live in forests might help us begin to construct a mental image of a bear in its natural environment before the word "forest" is encountered, or to interpret this same meaning despite a typographical error in the text such as "forrest." An interplay between bottom-up and top-down processes can build, refine, and adjust the situation model based on both exogenous and endogenous influences. This flexible, yet predictable, representation serves as a conceptual framework that makes it easier to comprehend, store, and retrieve information (Friston, 2005).

1.3 Neural Correlates of Knowledge Structure Building

Theoretical models describe largely consilient accounts of the cognitive construction of narrative representations. They consistently feature the construction of a global semantic representation that is iteratively updated as new information becomes available. From this firm foundation, we next review research that explores how cognitive models might be instantiated in brain-behavior relationships.

The *Structured Event Complex* (Wood & Grafman, 2003), specifies how knowledge macrostructures are stored in the brain, how they are activated or selected in response to neurological state changes, and how they enable prediction of future events. In a Structured Event Complex, prefrontal cortical "representations would store elements of knowledge…. Signified by the strength and pattern of neural activity" (Wood & Grafman, 2003, p. 2). The Structured Event Complex "represents thematic knowledge… concepts, social rules, and event features…. Aspects of [Structured Event Complexes] are represented independently but are encoded and retrieved as an episode" that is activated through spreading neural activation and "sustained firing of prefrontal cortical neurons" (Wood & Grafman, 2003, p. 4). Just as in PINS, this account of knowledge-structure building emphasizes the way smaller information units cohere to form a mental model, and that a state change occurs between the encoding of microstructural information and the attainment of macrostructural integrity.

Wolfgang Klimesch proposes a mechanism for the neurological state change that occurs as bottom-up and top-down knowledge processes interact (Klimesch, 2012) to achieve structural coherence. He describes access to the complex *Knowledge System* (an "internal," cognitive construct) as modulated by changes in alpha band neural oscillations; specifically, that alpha band desynchronization allows for access to the broad range of external sensory information necessary for bottom-up information processing, and selective attention to internally motivated tasks is maintained by high alpha power. (We will return and elaborate on the functional interpretation of alpha band activity in section 0.) To summarize, the Structured Event Complex of Wood & Grafman and Klimesch's Knowledge System describe the neural underpinnings of the cognitive mechanisms proposed by models of discourse representation. Cognitive science and neuropsychology thus converge in their accounts of flexible, stored representations of knowledge macrostructure in the human mind.

Because the processes of situation model construction and structure building are iterative and dynamic, neuroimaging techniques which can capture moment-to-moment changes are invaluable for providing insight into neurocognitive mechanisms. The precise temporal resolution of EEG allows for investigation of structure building processes as they unfold, which is not possible with techniques such as functional magnetic resonance imaging. In the next sections, we review the electrophysiological correlates of narrative comprehension, including a brief overview of event-related potentials and a more detailed discussion of oscillatory dynamics.

1.3.1. Event-Related Potentials

Event-related potentials (ERPs) represent the neural response to time-locked events from the scalp-recorded EEG. The N400 and P600 ERP components are associated with narrative- and discourse-level processing. Importantly, both are domain-general (i.e., are elicited by both verbal and visual stimuli; Coderre et al., 2018; Cohn, 2020b; Kutas & Federmeier, 2011; West & Holcomb, 2002) and thus thought to reflect broader cognitive functions involved in language comprehension.

The N400 is a negativity peaking at approximately 400 ms that has been broadly implicated in semantic processing (Bornkessel-Schlesewsky et al., 2022; Kutas & Federmeier, 2011), although debates remain regarding the specific level(s) of semantic processing that it reflects (Baggio, 2018; Brouwer et al., 2012; Kutas & Federmeier, 2011). The N400 attenuates across

sequence position in both sentences and sequential images (Coderre et al., 2018; Cohn et al., 2012; Coopmans & Cohn, 2022; van Petten & Kutas, 1991), which aligns with behavioral findings that participants spend longer reading initial positions of a discourse, or viewing panels in earlier sequence positions, than later ones (Cohn & Paczynski, 2013; Gernsbacher, 1985). This early-stage signal peak is thought to reflect the cognitive need to "lay a foundation" for mental representations (Gernsbacher, 1990). This effect may be general, reflecting that initial stages of semantic encoding might be more effortful than later (re)activation; or it may reflect the initial set-up of a narrative context followed by increasing ease of integration as more narrative information are presented and the situation model representation grows (Gernsbacher, 1990).

In contrast, the P600 is part of a group of inversely correlated (Aurnhammer et al., 2023) later positivities that are thought to reflect post-semantic integration and re-interpretation (van de Meerendonk et al., 2010). Like the N400, there are debates about which specific process(es) correspond with the P600 and other late positivities; still, they are generally thought to index backwards-looking updating, re-interpreting, or revising processes (Baggio, 2018; Brouwer et al., 2012; Kuperberg, 2021). Importantly, cognitive integration and revision are often associated with situation models and the iterative processes required for structure building.

The N400 and P600 components are thought to be inherently linked, with a larger N400 eliciting a smaller P600 and vice versa (Fromont, Royle, et al., 2020; Fromont, Steinhauer, et al., 2020; Kim et al., 2018; Pélissier, 2020; Tanner, 2019; Tanner et al., 2013). This has led some researchers to propose that the propensity for a larger N400 versus a larger P600 may reflect individual processing styles (Tanner & Van Hell, 2014).

1.3.2 Time-Frequency Analysis

In contrast to ERPs, which capture the brain's phase-locked neural responses at a specific time-point, *time-frequency analysis* (TFA) characterizes oscillatory neural dynamics across longer durations. For this reason, TFA is particularly advantageous for characterizing complex cognitive processes, such as discourse comprehension, which evolve over time (Gastaldon et al., 2020; Morales & Bowers, 2022; Schneider & Maguire, 2018). Oscillatory neural dynamics have been extensively studied and associated with various cognitive processes (Fries, 2015; Klimesch, 2012; Kosciessa et al., 2021; Meyer, 2018), and there is some evidence that neural dynamics in different frequency bands are organized with respect to the granularity and time-scale of the target information (Bai et al., 2022; Caucheteux et al., 2023; Meyer, 2018). Although there is still considerable debate about the roles of these different frequency bands in cognition (e.g., Woodman et al., 2022), we describe here some findings in the context of language and narrative comprehension.

1.3.2.1 Alpha

Oscillatory changes in the alpha (8-12 Hz) and beta (13-30 Hz) bands have primarily been implicated in attentional processes. Specifically, alpha power attenuation is thought to reflect the suppression of task-irrelevant information, as selected by top-down endogenous processes (Benedek et al., 2011; Dahl et al., 2022; Foxe & Snyder, 2011; Klimesch, 2012, 2018; Palva et al., 2011; Woodman et al., 2022). In visuospatial cueing paradigms, alpha power *increases* over sites ipsilateral to the cued direction (Foxe & Snyder, 2011), and *decreased* alpha power is positively correlated with the number of irrelevant distractors in a visual array (Woodman et al., 2022). In language paradigms, alpha power attenuation during complex sentence-level tasks

is associated with improved comprehension performance (Wang et al., 2021b), and alpha power attenuation is greater in the context of stronger conceptual relationships versus weak ones (Teige et al., 2018). Given the proposed roles of alpha in attentional processing and its demonstrated involvement in language processing, there is reason to expect that it may be modulated by structure building processes, especially those related to suppression.

1.3.2.2. Beta

Beta band activity has been implicated in prediction (Friston & Frith, 2015; Gastaldon et al., 2020; Loonis et al., 2017; Mamashli et al., 2019; Meyer, 2018), working memory (Mamashli et al., 2021), and representational set maintenance (Engel & Fries, 2010; Spitzer & Haegens, 2017). In the psycholinguistic literature, Rommers et al. (2017), Drijvers et al. (2018) and Terporten et al. (2019) support an interpretation of beta attenuation as related to alpha suppression, but employed at a higher level of language complexity or stimulus constraint. Baastiansen argues that low-beta modulation from 13-20 Hz may represent the highest level of mental model integration (Bastiaansen & Hagoort, 2006; Bastiaansen et al., 2010; Lewis et al., 2016). Similarly, Wang et al. (2018) describe the beta band as playing a role in the top-down process of "unification," i.e. integration of higher level knowledge-structures with attentionally directed lower level sensory input.

A growing body of research implicates beta power attenuation with the anticipation of structural boundaries or a change in cognitive set within predictable structures (Engel & Fries, 2010; Graber & Fujioka, 2020; Lewis et al., 2015). During the auditory presentation of spoken narratives, low-beta power at 14–21 Hz tracks with comprehension of propositional information, rather than acoustic or syllabic patterns, and increases across narrative time (Pefkou et al., 2017). Theoretical modeling in the auditory domain supports this interpretation that low-beta oscillations carry top-down signals associated with sequential macrostructures rather than relatively local patterns (Chao et al., 2022). Given these findings, we expect that beta band activity may also be modulated by structure building processes.

1.3.2.3. Alpha-Beta Coupling

There is broad agreement that beta power, like alpha, is related to top-down cognitive influences. However, while alpha suppression is thought to drive *inhibition* at the "stimulus" level (Foxe & Snyder, 2011), some suggest that beta power selectively *enhances* target signals at the level of the schema or mental model (Fries, 2015; Kosciessa et al., 2021). Whether in response to endogenous or exogenous influences (Meyer, 2018; Meyer et al., 2020), these accounts of alpha and beta band modulation are consistent with a dual mechanism of attentional mediation (Terporten et al., 2019): both alpha and beta modulation may serve to increase a target's signal to noise ratio in complimentary ways (Bornkessel-Schlesewsky et al., 2022; Mamashli et al., 2019).

This account of attentional control as differentially mediated by alpha and beta band oscillations is compatible with the bidirectional schema-structure-building mechanism specified by narrative comprehension theories (Cohn, 2020b; Gernsbacher, 1990; Kintsch, 1988). Under this interpretation, alpha suppression would inhibit information processing that is irrelevant to the emerging story structure, while beta suppression enhances or maintains activation of propositions relevant to the story's current mental representation. We thus have reason to expect that both alpha and beta frequency bands will show modulations during narrative

structure building.

1.3.2.4. Theta

Theta band (3-7 Hz) activity has been hypothesized to reflect *implementation* (Botvinick et al., 2004; Federmeier et al., 2020) of the attention-directed sampling of sensory stimuli, with modulation tracking bottom-up linguistic stimulus processing (Bai et al., 2022; Meyer, 2018; Pefkou et al., 2017; Poeppel & Assaneo, 2020; Strauß et al., 2014). Relatedly, theta has been associated with working memory or set maintenance (Engel & Fries, 2010; Prystauka & Lewis, 2019). For instance, during the comprehension of short verbal narratives, theta power increased in response to repeated names in comparison with new names (Coopmans & Nieuwland, 2020). The authors suggested that theta modulation was related to the activation of semantic information stored in working memory, and this is compatible with an interpretation that theta marks effective integration of microstructural information.

The theta band has also been implicated in conflict monitoring (Botvinick et al., 2004; Federmeier et al., 2020; Williams et al., 2021); this perspective may accommodate findings related to bottom-up activation of semantic information, as absence or resolution of perceptual conflicts permits set maintenance. Although the specific cognitive processes mediated by theta modulation during narrative comprehension has not been resolved, it is reasonable to expect that changes in theta-band power would be observed during narrative structure building, and may be associated with fine-grained stimulus driven processes.

1.3.2.5. Gamma

Frequently associated with theta dynamics (Buzsáki & Wang, 2012; Mellem et al., 2013), neural modulation in the gamma band (30-90 Hz) has also been associated with “semantic integration”(Bastiaansen & Hagoort, 2006; Bastiaansen et al., 2010; Coopmans & Nieuwland, 2020; Lewis et al., 2015). At the sentence level, gamma power is modulated by both semantic predictability and coherence (Wang et al., 2018), increasing in the context of information that coheres with the unfolding discourse, vs. incoherent information (Coopmans & Nieuwland, 2020; Murphy et al., 2023).

Gamma activity is thought to reflect bottom-up transmission of information within the constraints of top-down influence (Chen et al., 2026). In the context of predictive coding theory, Chao et al. (2022) found that gamma activation was associated with prediction-error signals, or conflicts between top-down representations and incoming sensory input. Taken all together, these studies suggest that gamma activity may be related to macrostructural coherence, and changes in gamma power observed during narrative structure building may be associated with the cognitive process of shifting or updating situation models.

1.4. The Current Study

This study is the first to explore the neural correlates of structure building across the entire trajectory of visual narrative comprehension, providing an important basis for understanding the emergent phenomenon of this cognitive process beyond the dynamics of its constituent parts. Using EEG, patterns of neurological activity were recorded as they changed over time in synchrony with the progressive building of narrative structure during graphic story comprehension.

1.4.1. Discourse Scale Analyses

Complex cognitive functions, such as narrative comprehension, rely on higher order dynamic neural processes (Friston et al., 2020; Gwilliams et al., 2025; MacWhinney, 2019; Owen et al., 2021). To understand how holistic narrative knowledge structures might emerge from sequentially presented constituent parts, a neurocognitive model must account for interactive, temporally dynamic processes that unfold across the timespan of the whole story. Current understanding of oscillatory dynamics, and alpha band correlates in particular, has been substantially influenced by studies focused on low-level sensory processing using canonical psychological tasks. For instance, many cueing paradigms ask participants to attend to target colors, shapes or spoken words in the context of distractors (Foxe & Snyder, 2011; Kosciessa et al., 2021; Woodman et al., 2022; Zhao & Woodman, 2020). Linguistic paradigms often prompt sentence completion at the word-level (Gastaldon et al., 2020) or track comprehension in the context of sentence-level syntactic variability (Terporten et al., 2019; Wang et al., 2021a). Our research builds on these studies to capture neural power dynamics that span narrative structure-building from beginning to end.

1.4.2. Visual Narratives

In the cognitive sciences, many explanatory models prioritize efficiency and adaptability by leaning away from domain specificity wherever possible. This study explores narrative comprehension in the visual domain to probe the extent to which empirical findings for verbally communicated stories are also evident using visual narratives. Lê et al. (2024) controlled for domain-specific effects in their study of working memory and inferencing during comprehension of verbal and visual narratives, finding that measures of working memory, widely understood as critical for discourse comprehension, were strongly correlated across the verbal and visual domains. However, it is difficult, if not impossible, to match the presentation of verbal and visual narratives in every relevant dimension simultaneously (Coderre et al., 2017; Coderre et al., 2018; Kubinski et al., 2025; Lê et al., 2023); nor should we expect such parity given their unique affordances (Cohn & Schilperoord, 2022a).

While written and pictorial stimuli are both appreciated visually, the linear sequencing required for reading differs from the appraisal of two-dimensional visual scenes (whether presented sequentially or not). These constraints would lead to expected differences in the temporal course of comprehension correlates. Relatedly, there are hemispheric biases in the processing of serial patterns presented at different time-scales (Kemmerer, 2022). Sensory-specific interactions should also be expected (Cohn, 2020b); early visual processing is localized posteriorly while early auditory processing is lateralized, and semantic knowledge is localized according to visual features in some cases, but not in others (Kemmerer, 2022). Therefore, we expect that neural correlates of core cognitive and executive functions, rather than stimulus-specific processes, will be similar across studies of narrative comprehension regardless of presentation modality.

1.4.3. Cognitive and Oscillatory Components of Structure-Building

Thus far, only one study has examined oscillatory power modulation during visual narrative comprehension: Coopmans & Cohn (2022) studied cataphoric and anaphoric referential

processing in visual narrative sequences using EEG. They observed differences in alpha (8-12 Hz) and gamma (45-65 Hz) power between cataphoric and anaphoric panels, and specifically, a greater decrease in alpha power for anaphoric panels compared to cataphoric panels. To date, no studies have examined the oscillatory dynamics of narrative structure building during the full time-course of visual narrative comprehension, nor linked changes in these dynamics to theoretical models of cognition.

Our exploratory investigation is informed by studies of populations with Autism (Scholten et al., 2021), traumatic brain injury (Lê & Coelho, 2023), schizophrenia (Gernsbacher et al., 1999), and attention-deficit-hyperactivity disorder (van Neste et al., 2015), which surface differences in long-range inferencing and macrostructural coherence. These aspects of structure-building facilitate the efficient encoding or retrieval of narratives, and can be subsumed under the umbrella of top-down cognitive processes. The current study explores the neural correlates of structure building, with a particular emphasis on top-down cognitive control. Based on our review of the literature, we examine alpha and beta power as our primary measures of cognitive structure-building. These frequencies have consistently been associated with top-down processes. In this way, our analysis has the potential to generate evidence consistent with alpha-beta modulation as a mechanism of cognitive control for structure-building in the context of narrative processing.

When interpreted in relation to models of narrative comprehension, increasing alpha-beta attenuation would be expected to track with the construction of the narrative schema, or knowledge superstructure as a unit. The suppression of irrelevant information as activation of a knowledge-structure simultaneously increases is conceptually consistent with narrative comprehension models like Gernsbacher's Structure Building Framework and Cohn's PINS model; Kintsch's Construction-Integration theory; and the prefrontal cortical neural activation of Structured Event Complexes described by Wood and Grafman.

Although we primarily expected changes in the alpha and beta bands, we also used TFA as a basis for data-driven exploration of change across the ranges of frequencies from theta to gamma. It is not clear from the literature how the many cognitive constructs implicated in effective structure building (Gernsbacher, 1991; Lê et al., 2012) would map onto the oscillatory dynamics of the five canonical frequency bands (Murphy, 2020); that task is too ambitious for this paper (Lindsay, 2020; Pickens et al., 2010; Wilhelm et al., 2013). Nevertheless, use of TFA permitted us to plot the correlates of narrative structure-building as the temporally continuous process that it is, providing us with a comprehensive, hypothesis-generating representation of the oscillatory dynamics underlying structure building during visual narrative comprehension.

2. Materials and Methods

For these analyses we used a dataset previously described in Coderre et al. (2020) and Kubinski et al. (2025). Those studies included data from both autistic and non-autistic participants; only data from non-autistic participants are analyzed here. All aspects of the study procedure were approved prior to the start of the experiment by the Institutional Review Board at the University of Vermont.

2.1. Participants

This study only included participants who reported neurotypical development and absence of any known neurological disorders or learning disabilities. We recruited 22 eligible local participants through newspaper, email list-serves, and flyer advertisements. They were all native English speakers. Their ages ranged from 18-50 (*M*=26, *SD*=9.3); 17 were female, and five were male. Two participants were left-handed, and all reported normal or corrected-to-normal vision. Informed consent was obtained from all participants according to the study procedures approved by the Institutional Review Board, and monetary compensation was provided in accordance with the same.

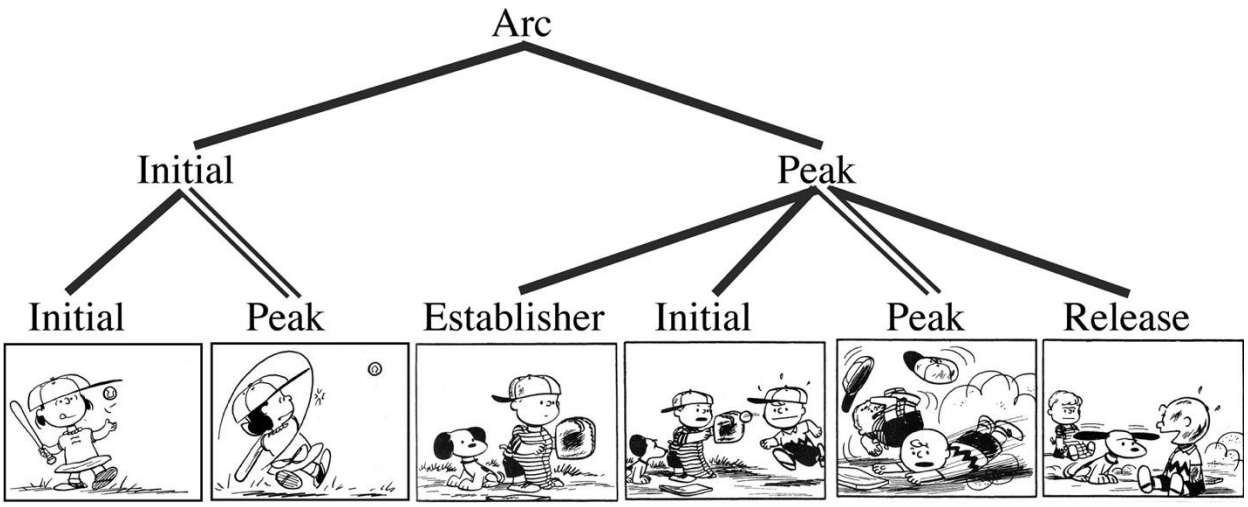
To quantitatively assess participants' expertise with visual narratives, we administered the Visual Language Fluency Index (VLFI; Cohn, 2014, 2020a). This standardized assessment is a questionnaire in which people report current and past experience with narrative comprehension in contexts such as text-only, movies, graphic novels, and comics. Participants earned VLFI scores ranging from 2 to 15, (*M* = 7). A VLFI score of 12 is considered "average" comic reading expertise, with <8 being considered "low" and >22 considered "high" visual language fluency. Part of the VLFI references general familiarity with *Peanuts* comic strips, and all participants in this study were familiar with *Peanuts*. This is relevant to our study, which uses visual features, characters, and themes that would be familiar to any casual reader of these comics. However, as described below, we constructed novel stimuli using panels from existing *Peanuts* comic strips, so no participant could be familiar with any specific sequence of narrative events presented during the experiment.

2.2 Stimuli

Participants viewed 120 six-panel wordless visual narrative sequences (comics). These stimuli were drawn from a larger corpus developed by recombining panels from Charles Schulz's *The Complete Peanut*s volumes into novel combinations. Each recombined narrative sequence was semantically and structurally coherent, and the "peak action" of the story arc occurred at either panel four or five. These sequences have been used in many studies of visual narrative processing, and most were originally validated for their comprehensibility in rating studies of both the whole sequence and adjacent panel pairs (see Cohn et al., 2012).

Just as verbal descriptions of initiating events, responses, attempts, and consequences fulfill superstructural requirements in Story Grammar analyses (Coelho, 2002; Merritt & Liles, 1987), visual elements in each panel signify their unique role in the narrative structure (Cohn, 2013; Cohn et al., 2012). According to Cohn's Visual Narrative Grammar (Cohn, 2013), these unique roles include: Establishers (which passively set up the scene and the relevant characters but do not include any active narrative action); Initials (which initiate a narrative action or event); Peaks (which depict the narrative climax or culmination of the action); and Releases (which resolve events). These components need not be linear in sequence, but can appear in more complex narrative structures depicting hierarchic constituencies, as shown in Figure 1.

Figure 1: Illustration of the narrative structure in an example sequence of images. Image from Cohn et al., 2012, Figure 1.



2.3 Procedure

Consenting, completion of behavioral assessments, and experimental testing all occurred during a single two-hour session. After signing a written consent form, each participant completed the VLFI questionnaire. Participants were then fit with a 128-channel EGI Geodesics Sensor net, which was linked to EGI's Geodesic EEG System (GES) 400 for amplification and signal capture. Next, participants were invited into a space that is visually partitioned from the experimenter and computers. They were seated 24" from the computer screen to approximate a viewing angle of 12.56° horizontally and 9.8° vertically when comic panels were presented at 5.25" wide and 4.125" high on our Dell 21.5" LCD monitor with 1920 x 1080 resolution.

Participants were instructed to attend to 180 narratives as they were presented in six blocks of 30; because one of the conditions used in the parent study (Coderre et al., 2020) is not relevant to our research question, we are reporting on the analysis of only 120 visual narratives.

Each six-panel sequence was presented one panel at a time using E-Prime version 2.0.10.356. Presentation was pseudorandomized across participants, who controlled the initiation of each narrative by pressing a button when they saw a "Ready?" screen. Each trial began with a 500 ms fixation cross, followed by each story panel visible in the center of the screen for 1350 ms with a 350 ms inter-stimulus interval.

To encourage attention to the narratives, comprehension checks were presented on screen after 20% of the trials. These were indicated by a red question mark above a comic panel with a red border. Participants pressed a button to indicate whether they thought the check-panel was related to the previous narrative sequence. A one-sample *t*-test confirmed that participants were significantly above chance (50%) on their responses to these comprehension questions ($t(43) = 25.76$, $p < <0.0011$; mean accuracy = 88%, $SD = 9.8$), indicating that they were paying attention to the narratives.

2.4 EEG Data Acquisition and Preprocessing

EEG data was collected using EGI's Geodesic EEG System (GES) 400, NetStation version 5.4, and a HydroCel 128-channel EEG net. Impedences were kept below 50 kΩ. EEG data were referenced to the Cz electrode and recorded continuously at 500 Hz, then subjected to a 4 kHz anti-aliasing lowpass filter.

We used the EEGLAB toolbox (version 14.1.1b; Delorme & Makeig, 2004) in the MATLAB (version 2022b) computing environment to preprocess and analyze the electrophysiological signal. Data was first filtered using a 0.1-50 Hz bandpass filter, then segmented into epochs time-locked to the onset of the first panel in each visual narrative sequence. Epochs extended from 700 ms before the first panel onset to 10,600 ms after; this long epoch duration was used to capture the full presentation of the 6-panel visual narrative sequence plus adequate pre- and post-stimulus time to allow for time-frequency decomposition at low frequency bands. Next, to prepare for artifact correction by independent component analysis (ICA; Delorme & Makeig, 2004; Jung et al., 2000), the mean of each trial was removed (Groppe et al., 2009) and data were reduced to 32 dimensions. We reviewed the ICA decomposition to identify components related to head movement, eye blinks, respiration, or other physiological artifacts; components that were obviously related to artifacts were removed from the sensor data by a trained experimenter. Segments were then baseline corrected using the first 100 ms of each epoch and re-referenced to the average of the mastoids. Unusable trials were initially identified and rejected using a joint probability computation (Delorme et al., 2007) with a threshold of 3 standard deviations; subsequent visual inspection of remaining trials ensured that all unusable trials were removed from the data.

2.5 Time-Frequency Analyses

We used the *newcrossf* function within EEGlab for coherence analysis, followed by spectral decomposition using a Morlet wavelet of 2 cycles with an expanding factor of 0.5 and a Hanning taper. Baseline correction of the power data was performed using data from the first 100 ms of each trial. We calculated power at nine electrodes: F3, Fz, F4, C3, Cz, C4, P3, Pz, and P4 based on the 10-20 electrode distribution. This electrode montage optimizes broad spatial representation (Coderre et al., 2017; Coderre et al., 2018) for this exploratory analysis. For each of the nine electrodes, we calculated power from 2-50 Hz at approximately every 10 ms. These parameters resulted in a window of time-frequency decomposition lasting from 143 ms prior, to 10,041 ms after the onset of the first panel in each visual narrative sequence. We analyzed spectral power in the theta (3.5-7.5 Hz), alpha (8-12 Hz), low beta (12.5-20 Hz), and gamma (30-50 Hz) frequency bands, with boundaries defined in alignment with previous work (Coben et al., 2008; Fries, 2015; Halgren et al., 2015; Lewis et al., 2016; Mellem et al., 2013; Meyer et al., 2015; Terporten et al., 2019; Wang et al., 2018).

2.6 Statistical Analyses

We performed a series of analyses to examine changes in power during visual narrative comprehension.

First, we visually examined the spectral plots of EEG power at every frequency (from 3-50 Hz, spanning theta, alpha, low beta, and gamma frequency bands. Next, to objectively identify

which frequency band(s) to target in subsequent statistical analyses, we performed mass univariate statistics (Groppe et al., 2011) to delineate when and where power deviated from baseline. One-sample *t*-tests (comparing power values to 0, which would indicate no change from baseline) were performed at each timepoint and frequency; correction for multiple comparisons was performed using a false discovery rate (FDR) correction. Mass univariate statistics were performed separately for each of the 9 electrodes, so we used a Bonferroni-corrected alpha level of 0.0056.

Next, we examined how power changes over the course of narrative processing by examining the slopes of power changes across panels at every frequency band. While examination of spectral power would show regions with significant power changes compared to baseline, it would not indicate whether power increases or decreases in a linear fashion over each subsequent panel of narrative comprehension. In this analysis, power across narrative time was plotted for every frequency in our spectral decomposition (1573 total). We added a line of best fit for each graph, then extracted the slope of that line. The slope values of the best-fit lines were then plotted for each frequency at each electrode.

While this analysis will tell us the slope of linear change in power over time, it will not provide inferential statistics to tell us whether this change in power is significant. Therefore, in our second analysis we statistically compared changes in power over time for each frequency band individually using linear mixed effects models. We calculated the average power in 100-ms time bins from 200-900 ms after each panel presentation at each of the nine electrodes across the scalp. (Note: Data coding issues resulted in missing data for panel 3 in the 900-1000 ms time window, therefore we only explored effects until 900 ms after each panel presentation.) In each time-bin we ran a linear mixed effects model on power in each frequency band in relation to fixed effects of *panel number* (1-6), *site* (frontal, central, parietal), and *laterality* (left, midline, right). Dummy coding was used for categorical variables, with the reference level for *panel number* being panel # 1, for *Site* being “frontal”, and for *Laterality* being “left”. No variables were scaled or transformed before analyses. *Subject* was included as a random effect (intercept only). *Item* could not be included as a random effect because the time-frequency decomposition assesses changes in power over all trials for each subject. We chose not to follow a maximal-to-minimal model selection process (Barr et al., 2013; Meteyard & Davies, 2020), such as exploring the inclusion of random slopes, because of the number of time windows and frequencies being analyzed in different models. Doing so could potentially generate different models, and different random effects structures, at each time window and/or frequency, which could complicate interpretation of results. Instead, we specified the same model every time, which is shown in Equation 1:

[1] *power ~ panel number * site * laterality + (1|subject)*

Because of the number of time bins (7) and frequency bands (4), we implemented a relatively strict correction for multiple comparisons, using a Bonferroni correction for 11 comparisons. This resulted in an alpha of 0.0045 for significance. All statistical analyses were performed using R version 4.4.1 (R Core Team, 2024).

3. Results

3.1. Visual inspection and definition of frequency bands

Figure 2 depicts changes in power across the frequency spectrum from 3.5-50 Hz, and over the full time-course of visual narrative presentation at all nine electrodes. Each spectral plot represents neural dynamics during the consecutive presentation of six-panel visual narratives, with time-frequency data for all participants' viewing of 180 narratives averaged together.

Visual inspection of Figure 2 suggests that both alpha and low-beta attenuation are minimal during the first panel and increase until panel 4. After panel 4, alpha and beta attenuation remain steady or slightly decrease. In addition to alpha and low-beta attenuation across narrative time, it is visually apparent that substantial power changes also occur in the frequency bands typically described as theta (3.5-7.5 Hz), and low-gamma (30-50 Hz). Moreover, the power trend in all four frequency bands appears to progressively decrease across narrative time. Visually perceptible regional differences include early theta power in posterior midline areas that decreases over time, decreasing gamma power that appears left lateralized in central regions, and low-beta suppression that is left lateralized in frontal areas.

Figure 2: Full spectrum power at all nine electrodes across narrative time. EEG power at the each of 9 electrode locations is averaged across all participants and all trials, and plotted over time beginning at the onset of the initial narrative panel presentation and lasting until the offset of the final narrative panel. Vertical lines indicate the onsets of each of the six panels in the sequence. Horizontal lines demarcate the boundaries between canonical frequency bands. Darker blue tones represent decreases in power.

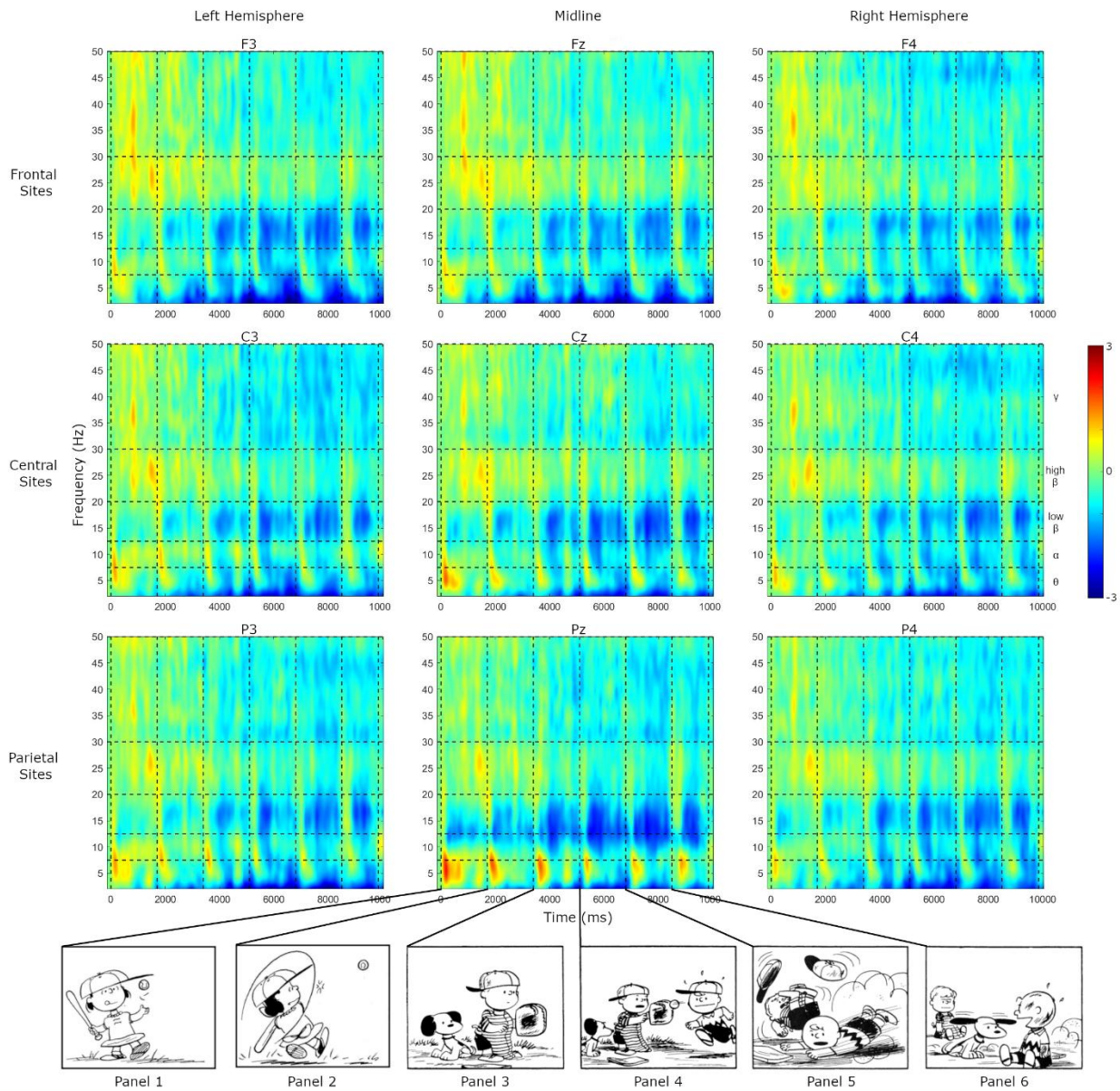
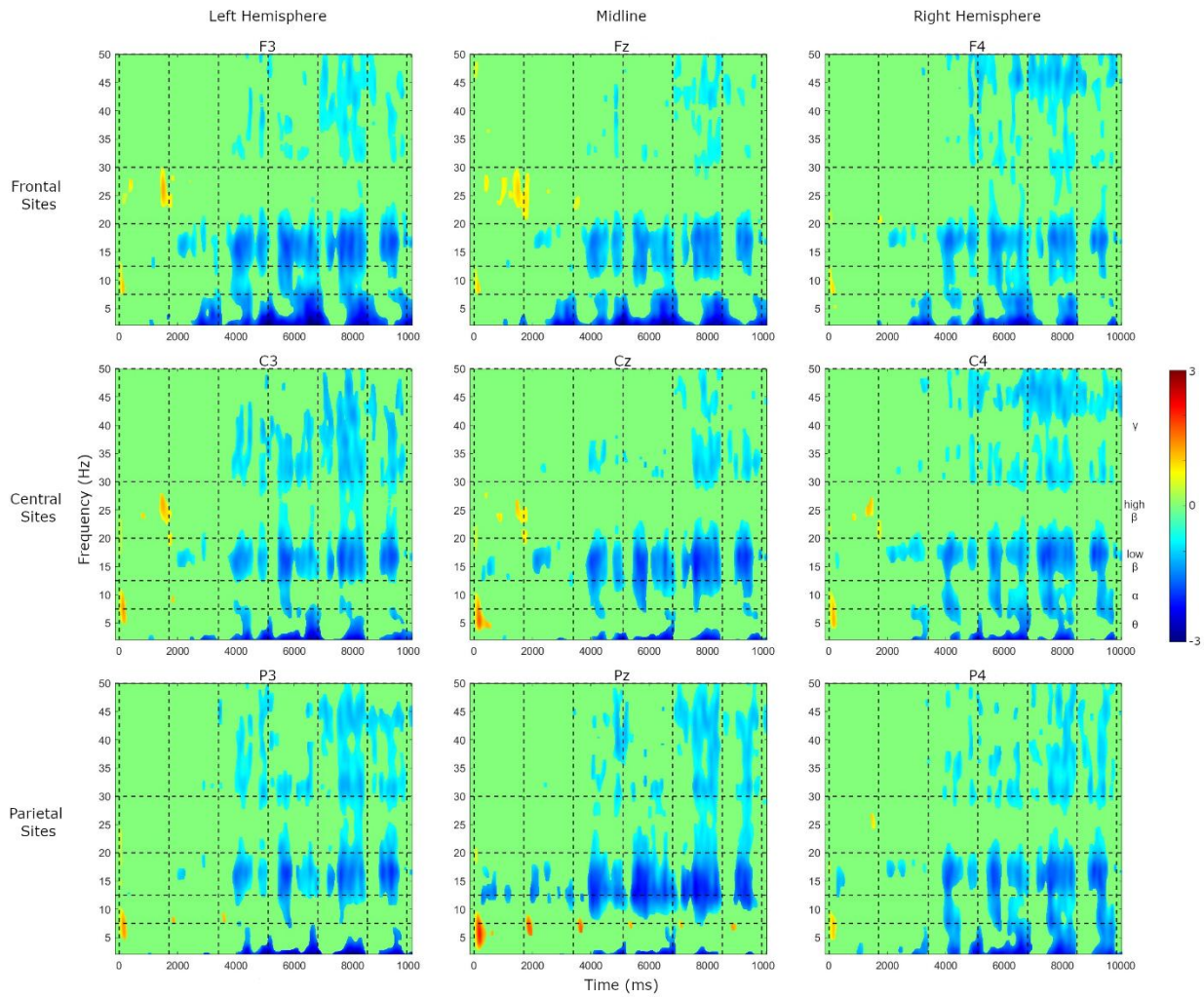


Figure 3 shows the spectral plots masked to exclude non-significant changes in power. Visual inspection of this masked data confirms our interpretation of Figure 2: there are significant power changes in the alpha band, theta band, low beta band, and gamma band. Accordingly, our subsequent analyses included all four of these frequency bands to explore the full extent of power changes during narrative comprehension.

Figure 3: Significant power changes at all nine electrodes across narrative time. Data points whose change in power from baseline were not significant (evaluated using mass univariate statistics with an FDR correction and a Bonferroni-corrected alpha level of 0.0056) have been masked and are shown in green.

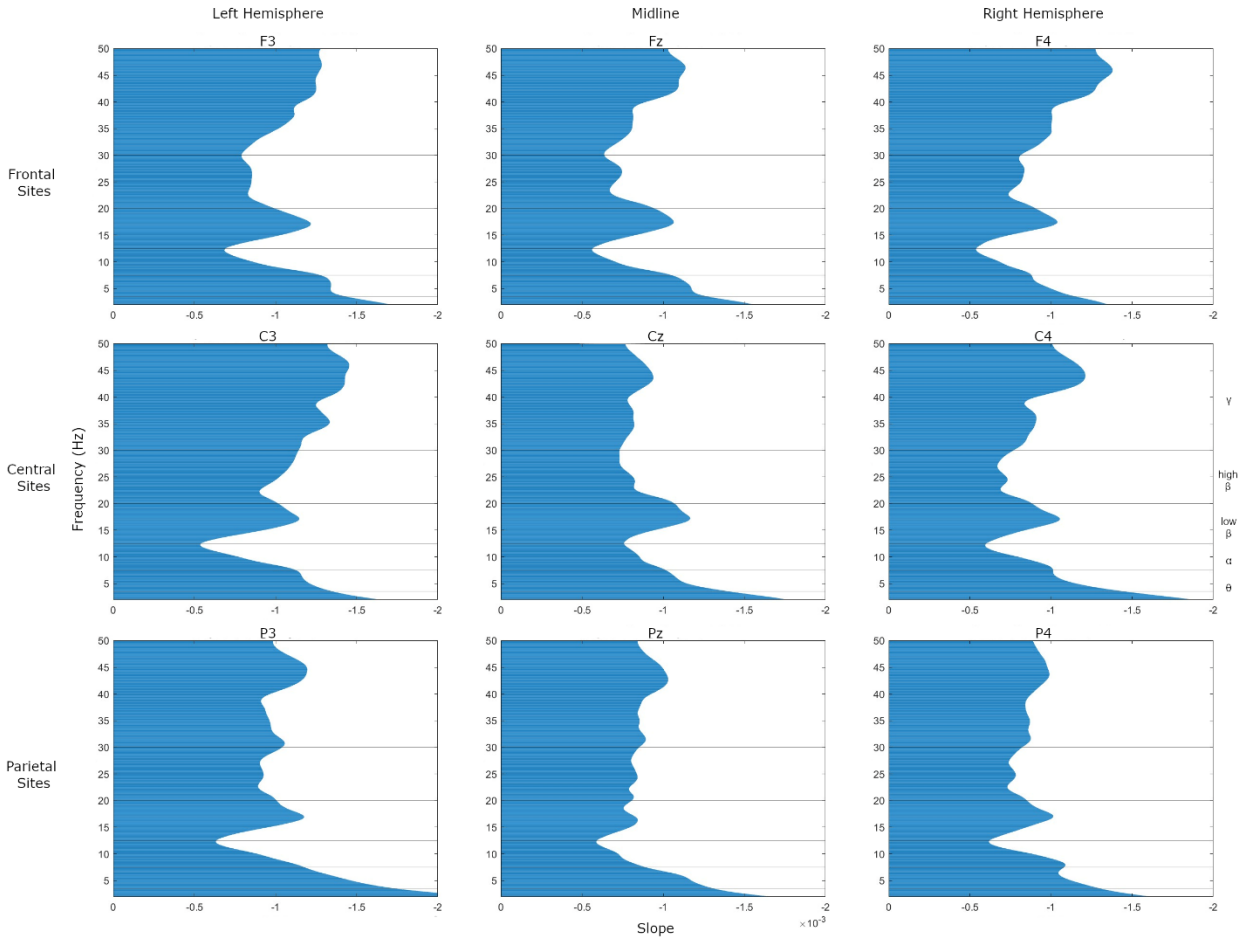


3.2. Slopes of best fit

To better visualize changes in power at each frequency across narrative time, we plotted the slope of best fit for power changes over the entire narrative. Figure 4 shows the slope of best fit plots at each of the 9 electrodes. Notably, all slopes are negative. This suggests that power gradually decreases over the course of narrative processing at every frequency band. However, there are notable differences among frequency bands in the strength of the power decrease over panel presentations. There appears to be a strong change in power in the low beta band, with a peak at approximately 17 Hz, which is more prominent at frontal and central sites than at parietal sites. Gamma also shows strong power decreases at frontal sites, while

power decreases in the lower alpha and theta bands (intensifying toward the delta range) dominate at posterior sites. Overall, this analysis supports the value in examining all frequency bands in subsequent analyses.

Figure 4: Slope of best fit at all frequencies across narrative time. Stronger values (positive or negative) indicate greater change in that frequency band over the course of the narrative.



3.3. Linear mixed effects models

Full results of the linear mixed effects models on the average power in each frequency band are provided in the Appendix. Table 1 provides the effects of panel number, which was the only factor which showed significant effects across any frequency band.

Table 1: *t*-values for the linear mixed effects models at each time window and frequency band. Only effects of panel number are shown, since this was the only factor which showed significant results; full model outputs can be found in the Appendix. Fixed effects in each model are *panel number* (1-6; panel 1 as reference level), *site* (frontal, central, parietal; frontal as reference level), and *laterality* (left, midline, right; left as reference level). *Subject* was included as a random effect. Statistically significant effects of *panel number* (using a Bonferroni correction of 0.0045 to account for the 7 time windows plus 4 frequency bands) are highlighted in bold.

FREQUENCY BAND	PANEL CONTRAST	200-300 MS	300-400 MS	400-500 MS	500-600 MS	600-700 MS	700-800 MS	800-900 MS
THETA	(Intercept)	0.89	1.01	1.20	0.83	0.59	0.16	-0.36
	panel number (1 vs 2)	-0.82	-0.81	-1.06	-1.22	-1.30	-1.00	-0.74
	panel number (1 vs 3)	-1.37	-1.81	-2.28	-2.52	-2.69	-2.43	-2.04
	panel number (1 vs 4)	-2.82	-2.43	-2.62	-2.42	-2.41	-2.39	-1.85
	panel number (1 vs 5)	-2.62	-1.79	-1.74	-1.37	-1.35	-1.57	-1.29
	panel number (1 vs 6)	**-3.34**	-2.27	-2.18	-1.97	-1.83	-1.40	-1.25
ALPHA	(Intercept)	0.05	-0.44	0.04	0.11	0.13	0.23	0.45
	panel number (1 vs 2)	0.33	-0.03	-0.87	-1.36	-1.18	-0.96	-1.05
	panel number (1 vs 3)	0.35	-0.85	-2.14	-2.72	-2.61	-2.56	-2.36
	panel number (1 vs 4)	0.03	-1.42	**-3.01**	**-3.07**	**-2.93**	**-3.06**	-2.38
	panel number (1 vs 5)	-0.07	-0.27	-1.78	-1.86	-1.57	-2.40	-2.77
	panel number (1 vs 6)	-0.05	0.55	-1.19	-1.71	-1.94	-2.29	-2.30
LOW BETA	(Intercept)	-0.51	-0.98	-1.19	-1.53	-1.12	-0.60	-0.24
	panel number (1 vs 2)	-0.91	-1.18	-1.40	-0.94	-0.82	-1.34	-1.79
	panel number (1 vs 3)	-0.91	-1.26	-1.59	-1.72	-2.82	**-4.08**	**-3.46**
	panel number (1 vs 4)	-0.37	-1.94	**-3.14**	-2.79	**-3.17**	**-4.10**	**-3.26**
	panel number (1 vs 5)	-0.93	-1.98	-2.61	-1.69	-1.79	**-4.04**	**-4.24**
	panel number (1 vs 6)	0.21	-0.06	-1.80	-2.36	-2.67	**-4.04**	**-3.86**
GAMMA	(Intercept)	1.39	1.54	1.55	1.59	1.29	1.98	2.24
	panel number (1 vs 2)	-1.06	-1.05	-0.44	-0.92	-0.82	-1.06	-1.52
	panel number (1 vs 3)	-2.16	-2.32	-1.84	-2.06	-2.19	**-3.29**	**-2.85**
	panel number (1 vs 4)	-2.30	-2.63	-2.80	**-3.19**	**-2.90**	**-3.53**	**-3.06**
	panel number (1 vs 5)	**-3.73**	**-3.54**	**-3.31**	**-3.00**	-2.53	**-3.64**	**-3.63**
	panel number (1 vs 6)	**-3.41**	-2.52	-2.65	-2.64	-2.31	**-3.42**	**-3.52**

In the theta band, the only effect of panel number occurred between panels 1 and 6 from 200-300 ms. To visualize these effects, we plotted the average theta power over every panel in this time window; the results are shown in Figure 5

Figure 5a. As can be seen there, theta power decreased steadily from panels 1 to 6, with a slight plateau between panels 4 and 5.

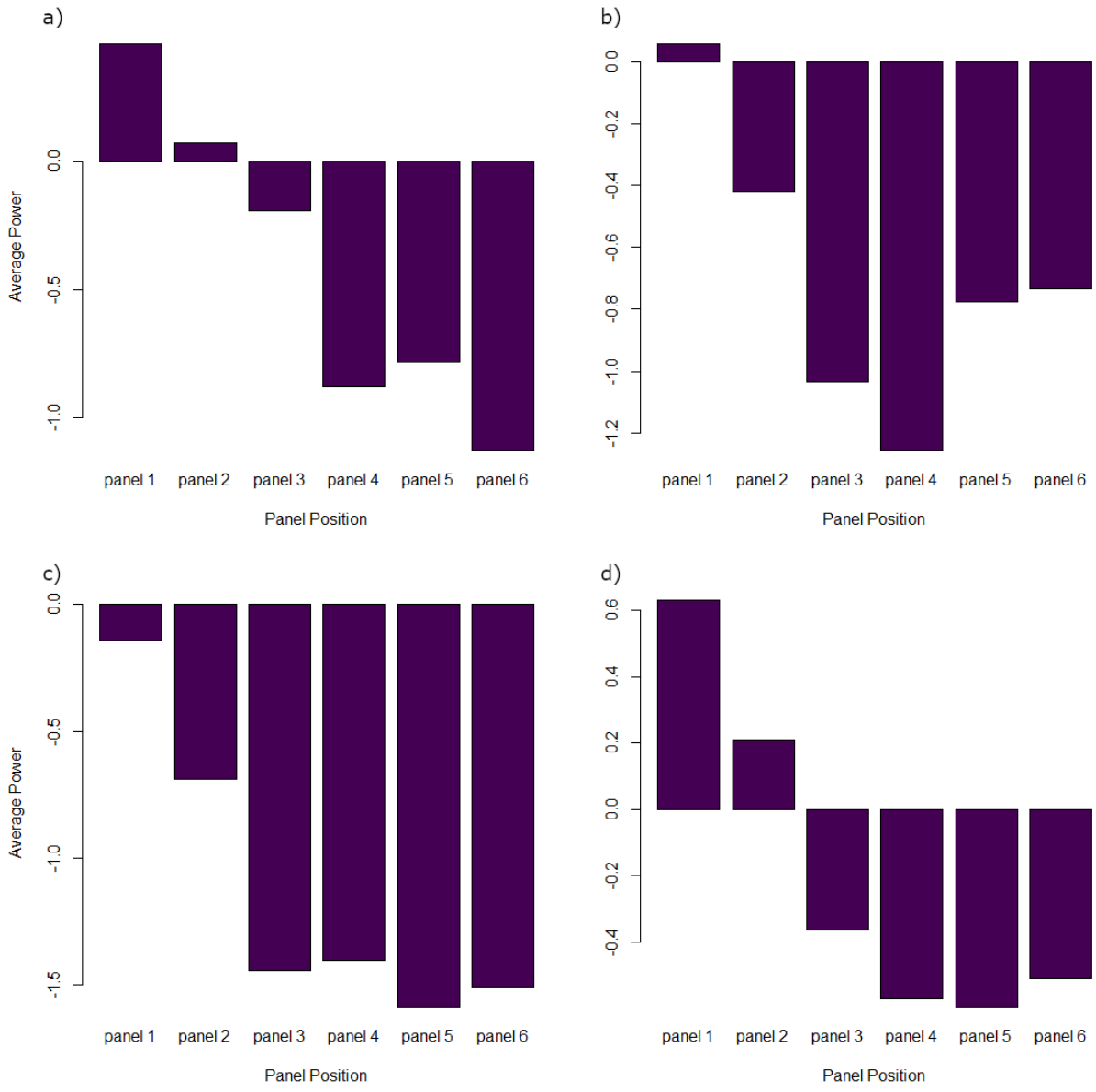
Alpha power showed an effect of panel number, specifically in the contrast between panels 1 and 4, from 400-800 ms. To visualize these effects, we collapsed power over this entire time window and plotted average alpha power; the results are shown in Figure 5

Figure 5b. As can be seen there, alpha power decreased steadily from panels 1 to 4, before increasing in panels 5 and 6, respectively.

Low beta power showed a reliable effect of panel number from 700-900 ms, with significant differences between panels 1 and panels 3-6. Differences between panels 1 and 4 also occurred from 400-500 ms and from 600-700 ms. To visualize these effects, we collapsed power over the time window from 700-900 ms and plotted average low beta power; the results are shown in Figure 5c. As can be seen there, low beta power decreased from panels 1 to 3, plateaued slightly, then decreased slightly more at panels 5 and 6.

Gamma power showed a significant effect of panel number from 200-900 ms, particularly between panels 1 vs. 4 and 1 vs. 5. To visualize these effects, we collapsed power over the time window from 500-900 ms and plotted average gamma power; the results are shown in Figure 5d. As can be seen there, gamma power decreased steadily from panels 1 to 5, then increased slightly at panel 6.

Figure 5: (a) Average theta power from 200-300 ms after each panel. (b) Average alpha power from 400-800 ms after each panel. (c) Average low beta power from 700-900 ms after each panel. (d) Average gamma power from 500-900 ms after each panel. Averages were performed at the reference electrode (left frontal site, as used in the dummy coding in the statistical models) for each frequency.



4. Discussion

This study explored the time-frequency signatures of structure building during narrative comprehension. We examined EEG data in response to visual narratives, and we believe that our results would also be applicable to stories told through other modalities, given the domain

generality of structure-building processes. Although we primarily expected to find that alpha and low beta power would track with the incremental establishment of narrative structure, due to the exploratory nature of this study we also examined other frequency bands in lower (theta) and higher (gamma) ranges. Our analyses revealed significant, non-linear power changes in in four canonical frequency bands over the duration of narrative time; these trajectories are consistent with theoretical accounts of narrative structure-building.

In the theta band (3.5-7.5 Hz), we observed significant power deviations from baseline in an early time window: 200-300 ms. These early effects that are stable across panels are consistent with interpretations that theta band activity represents the bottom-up tracking or processing of relatively granular exogenous stimuli (Fries, 2015; Meyer, 2018; Prystauka & Lewis, 2019). Because the process of visual appraisal should be relatively similar at every panel, the lack of a substantial modulation in theta power by panel number is not surprising.

In the alpha band (7.5-12 Hz), we observed significant changes in power over the course of narrative comprehension such that alpha power decreased gradually from panel 1 to panel 4 before beginning to increase again in panels 5 and 6 (although still maintaining an overall decrease in these latter panels relative to baseline). The low beta band (12.5-20 Hz) showed a slightly more linear trajectory, with power gradually decreasing from panel 1 to panel 3, then plateauing at panel 4 before increasing slightly again in panels 5 and 6. We elaborate on potential interpretations of power changes in these frequency bands in the next section.

Finally, in the gamma band (30-50 Hz) we observed notable change in power over narrative time. We would interpret decreasing gamma power over narrative time as related to the increasing reach and strength of the narrative situation model (Bastiaansen & Hagoort, 2015; Meyer, 2018; Prystauka & Lewis, 2019). During early structure building, gamma power may signal new propositions being incorporated into the growing situation model; in later stages of narrative comprehension, the active schema increasingly constrains narrative expectations, so the neural effects of structural additions and revisions are less prominent. This interpretation is in alignment with research showing decreased pre-frontal activation in the context of well-structured narratives (Cannizzaro et al., 2012).

4.1. The Roles of Alpha and Beta

During narrative comprehension, we primarily expected to see changes in the alpha and beta bands; our data-driven exploratory analyses confirmed that these were the two frequency bands most significantly modulated during narrative processing. Here we devote some additional discussion to what our observed findings in these frequency bands may represent.

In the alpha band, we observed gradually decreasing alpha power as the narrative unfolded. Some studies have found that alpha power attenuation is larger in the presence of distracting information (Woodman et al., 2022). In the context of our study, this decrease in alpha power could reflect the ongoing suppression of information that is not related to the unfolding situation model. For instance, as a visual narrative unfolds, each panel shows information that is essential to the plot as well as details that are extraneous. In understanding the story, the comprehender will need to identify which details are relevant, and incorporate them into the ongoing situation model, as well as discard information that is not necessary. The decreases in alpha power that we observe may reflect this process of selecting what is relevant and discarding what is not. Under this interpretation, it would make sense that alpha power

attenuation would increase as story relevance becomes more apparent: in the early panels, it is not yet known what the story is "about;" as events unfold in subsequent panels, the "gist" of the story becomes more clear, making it easier to identify and discard narrative-irrelevant information. Under this interpretation, the significant panel effects in our study suggest increasing top-down control of attention over narrative time.

This speculative proposal aligns with an aspect of Gernsbacher's Structure Building Framework (Gernsbacher, 1991), described briefly in section 1.1, that she calls "mechanisms of suppression and enhancement." Gernsbacher proposes that memory nodes make up the mental representation of the narrative, or the situation model, and that activation of those memory nodes is controlled by enhancement and suppression mechanisms. These mechanisms can explain why comprehenders typically lose access to "surface" information in a narrative (i.e., details that do not contribute to meaning) while "thematic" information (the overall "gist") is maintained: Surface level information changes rapidly with incoming stimuli, meaning that old surface level information must be suppressed in order to process the new. In contrast, thematic information is enhanced with repetition during narrative development, so contributes to the construction of the situation model. In our data, alpha power attenuation over narrative time could reflect an increasing need to *suppress* surface information and enhance thematic information.

Similar patterns of power changes occurred in the low beta band, with beta incrementally decreasing from panel 1 to panel 4. These patterns support our expectations of seeing low-beta power decrease as a situation model is incrementally constructed. This frequency band, however, also showed regional differences between sites which were not observed in the alpha band. Specifically, the decrease in beta power became significantly different from panel 1 at an earlier panel (#3) at frontal sites than at central and parietal sites (#4) and compared to alpha (#5 at all sites). This earlier onset of beta attenuation may be compatible with the idea that beta influences selective signal *enhancement* even in the context of relative uncertainty (e.g., early in the narrative while the plot is still developing). We contrast this with the explanation that more global alpha attenuation has a later onset and peak (panel #5) that is delayed until the story schema has been well developed and stimulus uncertainty reduced.

In this context, our results provide evidence for a model of visual narrative comprehension that utilizes alpha-beta modulation to balance comprehension accuracy with cognitive efficiency. Through the lens of structure building, a dynamic balance of bottom-up and top-down processes may be used to create narrative coherence in mental models. This interpretation of our findings is speculative, especially given that our analyses are exploratory and we did not experimentally manipulate our stimuli. To test our interpretation of oscillatory power dynamics during narrative comprehension, future work should explore the relationship of predictability and cognitive efficiency by actively varying the need for and ease of identifying irrelevant information in a narrative. We offer some suggestions for such future directions in the next section.

4.2. Limitations

As this work is exploratory, there are some limitations to mention. The use of many time windows for statistical analysis permits a comprehensive view of significant differences across temporal power trajectories. While our analysis reveals an integrated picture of this variability

over time, we examined each frequency band in isolation, which limits our understanding of the potential interactions posited by models that invoke cross-frequency coupling (Pagnotta et al., 2024; Palva & Palva, 2018; Siebenhühner et al., 2016). We were also not able to explore the delta band (approximately 1-4 Hz) given the limitations of the duration of our EEG segments. Future work should prioritize a full exploration of slower-wave oscillations and the use of analytic tools that better capture the complexity of signal variability, including quantifying the interaction between frequency bands over time.

Relatedly, this proof-of-concept study used an existing dataset to explore time-frequency signatures of narrative comprehension; however, secondary data analysis doesn't allow for experimental manipulation to precisely target research questions. An important future step will be to systematically manipulate narrative stimuli to test predictions about how power in different frequency bands will change. These manipulations should target narrative features likely to affect the structure building process. For instance, our analysis reveals differentiation between alpha and beta trajectories, but does not shed light on the nature of this distinction.

Another limitation of the current study is that we only explored narrative structure building in the visual modality. While the theoretical perspective of domain-general linguistic cognition is increasingly supported in the literature (Cohn & Schilperoord, 2022b; Federmeier et al., 2020; Fedorenko et al., 2024; Hodgson et al., 2021; Lê et al., 2023), few studies have tested this domain generality by directly comparing visual and verbal modalities. Although we theorize that similar neural mechanisms, most notably a decrease in alpha and low beta power over narrative unfolding, would be employed for both visual and verbal narratives, there have been no studies thus far directly comparing modalities.

4.3. Future Directions

The current study explores the neural correlates of structure building, with a particular emphasis on top-down cognitive control. It will be important to investigate how individual differences in cognitive skills could affect narrative structure building (McDaniel et al., 2021). For instance, differences in fluency, relating both to reading comprehension skills in the verbal domain and to visual language fluency in the visual domain, could also affect the ease and efficiency of structure building processes. Fluency and age of acquisition are known to modulate N400 and P600 ERP responses during both language processing and visual narrative comprehension (Coderre & Cohn, 2023; Tanner et al., 2014). The extent to which these variables also influence oscillatory dynamics during narrative comprehension remains unknown.

Predictability may also modulate alpha and beta power dynamics during narrative comprehension. For predictable, highly familiar story types, increased top-down processing could maximize comprehension speed and efficiency; selective attention to a confirmed schema could inhibit the active processing of irrelevant details through alpha suppression, while beta suppression would boost the saliency of target information. Conversely, there may be increased need for computationally costly distributed attention throughout the unfolding of an atypical narrative to facilitate accurate identification of potentially significant changes, patterns, or pieces of information which can serve to continually update a lean mental schema. The contrast between cognitive processes that mediate comprehension accuracy and efficiency may be important for understanding structure building profiles across the spectrum of

individual and clinical differences.

Characterization of alpha and beta dynamics during experimentally controlled circumstances could clarify whether individual differences in comprehension are indeed related to inefficient suppression mechanisms (as potentially measured by alpha band synchrony), impaired activation and maintenance of structural representations, or other factors related to the narrative structure itself.

5. Conclusions

This research is the first to plot the discourse processing of visual stories across their entire trajectory. Using time-frequency analyses of EEG data during the full time-course of visual narrative comprehension, we observed non-linear power trajectories in four canonical frequency bands during visual narrative comprehension. Together, these dynamic trajectories describe a neural signature of narrative structure-building and situation model construction that is more complex than would be expected for sequential effects alone.

We observed that alpha and low-beta frequencies have a power trajectory that dips with statistical significance near panels 4 and 5 during narrative structure-building. This provides evidence that narrative structure-building does not merely induce a sequential, or additive panel-by panel effect of increasing working-memory demands over time. Similarly, theta and gamma power does not change linearly across the duration of narrative comprehension, but follows trajectories that are compatible with their roles as specified by theories of narrative structure-building: namely, that the theta rhythm is stimulus driven and would not follow a global narrative arc, while gamma would track with propositional integration and revision at the level of schema or discourse.

Overall, the neural dynamics observed in this study may be consistent with the incremental establishment of a global mental model, where cortical activation is attenuated over time as comprehension is scaffolded by a coherent narrative structure.

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